

Twin Rollouts: Noise-Coupled Counterfactual Branching in Interactive Video World Models

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Abstract

Interactive video world models generate rollouts autoregressively under an action stream, yet they are trained and evaluated almost exclusively on factual prediction. We study counterfactual generation *inside* the rollout: given a trajectory the model has itself generated, what would have happened had the actions differed from step t^* onward? We formalize **noise-coupled twin rollouts** — a factual and a counterfactual branch sharing the generated prefix and the future exogenous noise sequence, diverging only in the action stream at an intervention point. Because the factual branch is self-generated, its exogenous noise is known exactly: the abduction step of Pearl’s counterfactual procedure is **exact by construction**, sidestepping the approximate-inversion problem faced by editing-based pipelines. Noise coupling further turns the minimal-change principle into a **per-sample verifiable property**: we define a spatiotemporal locality metric that penalizes divergence outside the causal descendants of the intervention, computable against simulator ground truth without a learned judge. Forking the simulator state at t^* yields ground-truth counterfactual re-renders, which we use as **verifiable rewards** for post-training. This note establishes the formal framework, metric definitions, and positioning; experiments are forthcoming.

1 Introduction

Interactive video world models roll forward autoregressively under a stream of actions. Their training objectives, and nearly all of their evaluations, concern the *factual* rollout: how faithfully the model continues the trajectory it is on. This note concerns a different question, asked from inside the rollout: had the actions differed from some step t^* onward, what would this very trajectory have become? We make three contributions.

(C1) In-rollout counterfactual branching. To our knowledge, we give the first formalization of *generation-time counterfactual branching* for autoregressive interactive video world models: counterfactual branches share the self-generated factual prefix and the future exogenous noise sequence, diverging only in actions from an intervention point t^* . The construction is **abduction-free**: unlike editing-based counterfactuals that approximately invert an observed video, the exogenous noise of a self-generated trajectory is stored, not inferred.

(C2) Per-sample verifiable locality. Under noise coupling, any factual–counterfactual divergence outside the causal descendants of the intervention is attributable to a locality violation rather than to sampling stochasticity. We define a spatiotemporal locality metric using simulator-provided descendant masks: divergence must (i) begin only at t^* and (ii) remain confined to the descendant region as it propagates. The metric requires no learned judge and is directly reusable as a training signal.

(C3) Counterfactual-pair verifiable rewards. Forking the simulator state at t^* yields ground-truth counterfactual re-renders. We combine counterfactual outcome fidelity (against the forked ground truth) with locality (C2) into verifiable rewards for RL post-training — to our knowledge the first post-training objective that directly optimizes the *counterfactual correctness* of a video world model, as opposed to reconstruction, perceptual, geometric, or action-following rewards.

Separability. These contributions are separable: the locality metric (C2) and the counterfactual-reward objective (C3) do not presuppose the branching formalism (C1) and stand on their own.

2 Formal framework

Definition 1 (Interactive rollout with explicit noise). A world model generates $x_{k+1} = G_\theta(x_{\leq k}, a_k, \varepsilon_k)$, where ε_k collects all exogenous randomness consumed at step k — for diffusion/flow backbones, the initial latent noise and any stochastic-solver noise of chunk k ; for autoregressive token backbones, the per-position sampling variates (inverse-CDF uniforms or, equivalently, Gumbel keys) of the tokens of step k . A rollout $\tau = (x_0, a_{0:T-1}, \varepsilon_{0:T-1}, x_{1:T})$ is fully determined by (x_0, a, ε) given θ .

Definition 2 (In-rollout intervention; noise-coupled twin). Given a factual rollout τ^F and $t^* \in \{1, \dots, T-1\}$, an intervention is $I = \text{do}(a_{t^*:T-1} := a'_{t^*:T-1})$ (action edit) or $\text{do}(x_{t^*} := x')$ (state edit). The noise-coupled counterfactual branch is $\tau^{CF} = \text{Roll}_\theta(x_{\leq t^*}^F, a', \varepsilon_{t^*:T-1}^F)$: same prefix, same future noise, different actions. Exogenous noise is indexed by absolute timestep: the counterfactual branch consumes ε_t^F at world time t regardless of its own generation index.

Definition 3 (Verifiable spatiotemporal locality & outcome fidelity). Forking the simulator at t^* under I yields the ground-truth counterfactual render y^{CF} and per-frame causal-descendant masks D_t . Define

$$\begin{aligned} \text{locality violation } L &= \sum_{t > t^*} d(x_t^{CF} \odot (1 - D_t), x_t^F \odot (1 - D_t)), \\ \text{outcome fidelity } O &= \sum_{t > t^*} d(x_t^{CF} \odot D_t, y_t^{CF} \odot D_t), \\ \text{reward } R &= -(\lambda_O \cdot O + \lambda_L \cdot L). \end{aligned}$$

Both terms are computable without a learned judge.

Remark (exact abduction). Since τ^F is generated by the model, ε^F is stored rather than inferred; Pearl’s abduction step is exact. In contrast, inversion-based pipelines recover exogenous noise only approximately, and identifiability of exogenous noise in high-dimensional generative models remains open [8]. Noise coupling is what converts minimal-change from a distributional criterion into a per-sample testable one.

Remark (operationalization). D_t is computed from the paired ground-truth branches on the channels the simulator exports: the per-pixel difference channel always, and an entity-level channel where stable entity identities are available. On frames where the complement of D_t is empty — in particular under interventions whose viewpoint change covers the frame — L is *undefined rather than zero*, and we therefore report L stratified by intervention class. Pixel

comparisons exclude fixed overlay regions and apply a small magnitude threshold to absorb characterized renderer noise, which is attributable to neither a locality violation nor sampling stochasticity.

3 A minimal illustration

Figure 1 instantiates the construction in a small deterministic environment as an existence proof: the twin branches are generated by a trained autoregressive world model from a shared prefix and shared per-position sampling variates; the descendant masks come from the simulator fork. With coupling on, the region outside D_t is nearly silent, and what remains is a genuine locality violation of the model. With coupling off, the same region lights up with sampling stochasticity, and the per-sample attribution of Definition 3 is no longer available. Scale-up to a full interactive video world model, together with the data engine and post-training results, is deferred to the next revision.

4 Positioning

vs. CWMDT. CWMDT [6] formalizes counterfactual world models that accept interventions as explicit inputs, conditioning a video diffusion model on LLM-edited digital-twin text derived from an *observed* scene. This is an open-loop, single-shot setting: the factual video is externally given, interventions are semantic attribute edits, and generation occurs once. We study the complementary *closed-loop autoregressive* setting: the factual trajectory is generated by the model itself under an action stream, interventions are applied *mid-rollout* to actions or state, and the factual/counterfactual branches are coupled through shared future exogenous noise — making abduction exact rather than approximated through an inversion or reasoning pipeline.

vs. What-If World. What-If World [2] contributes an evaluation benchmark for *contrastive intervention following*: paired test instances anchored on real frames, differing in a single input-level variable, scored by the APEO protocol over black-box generations. Our work differs on three axes: (i) we contribute a generation method and a training objective, not an evaluation suite; (ii) our interventions occur mid-rollout on the action stream rather than at the input; (iii) our locality metric is defined under *controlled exogenous noise* — a condition unenforceable when evaluating black-box models — which converts minimal-change from a distributional criterion into a per-sample verifiable one.

vs. image-domain counterfactual diffusion and CSVC. A mature line of work instantiates Pearl’s abduction–action–prediction procedure in diffusion models via DDIM inversion for images [5, 4, 8], and via prompt-steered black-box editing for observed videos (CSVC [7]). We claim no novelty for this correspondence. Our departure is the setting: no observed artifact is inverted; the world model’s own rollout supplies the factual branch with exactly known noise, and the counterfactual is a *branch of the same generative process*, not an edit of a given video.

vs. PersistWorld. PersistWorld [1] branches multiple candidate continuations from a frozen prefix to stabilize RL post-training; its branches share the *action* sequence and differ in sampling stochasticity. Our construction is the exact dual: branches share the *noise* sequence and differ in

actions. The former averages over noise to stabilize training; the latter controls noise to expose the causal effect of actions.

vs. the counterfactual-controllability framework. Concurrent work [9] articulates counterfactual controllability as a design criterion for self-evolving world models at the framework level, without a concrete generation or training procedure. We provide, to our knowledge, the first method-level instantiation with a verifiable training objective.

vs. CounterScene. CounterScene [3] performs counterfactual guidance for safety-critical scenario generation in driving-specific BEV world models with predefined agent-interaction graphs. Our formulation is domain-general over interactive video world models and requires no hand-specified interaction structure; descendant sets are obtained from simulator ground truth.

vs. the RLVR-for-world-models line. RL post-training of video world models with verifiable rewards is established [10], but existing rewards target reconstruction, perceptual similarity, geometric consistency, or action following — all properties of a *single factual* rollout. Ours is, to our knowledge, the first reward defined on *factual-counterfactual pairs*, directly optimizing counterfactual correctness and locality.

5 Limitations

The locality metric inherits the capabilities of the simulator that grounds it. Under egocentric cameras, interventions that move the viewpoint drive the pixel-difference channel of D_t toward full-frame coverage, rendering L vacuous for that intervention class; we therefore stratify all locality reporting by intervention class, treat empty-complement frames as undefined, and confine training signal to viewpoint-preserving interventions until camera-compensated comparison is validated. Renderer nondeterminism, where present, is handled by fixed-region masking and magnitude thresholds rather than assumed away. The illustration in Figure 1 is an existence proof in a deliberately small environment, not evidence of scale.

References

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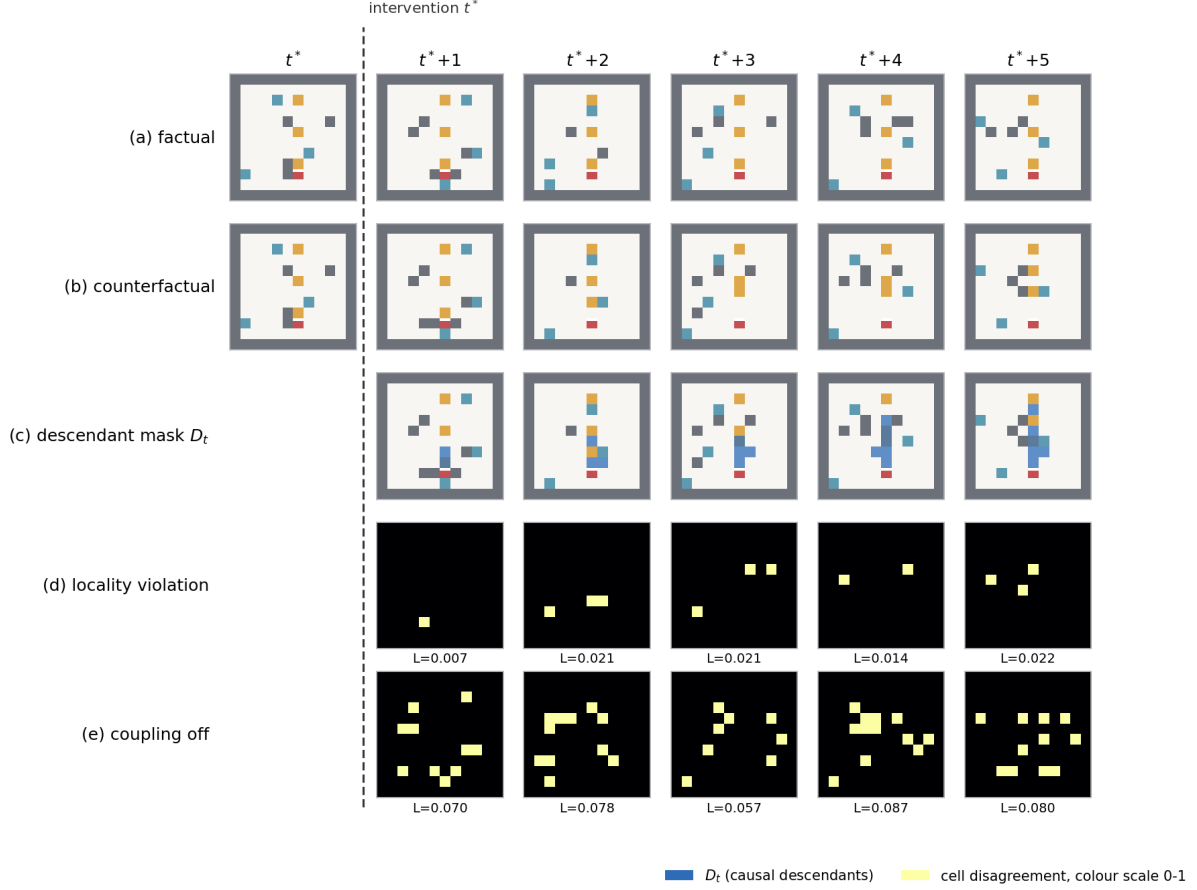


Figure 1: Noise-coupled twin rollouts in an interactive world model, instantiated in a deterministic grid world whose descendant masks are exact by construction. Rows (a) and (b) share the self-generated prefix and the future exogenous noise sequence, diverging only in the action stream from the intervention point t^* . Row (c) shows the causal-descendant mask D_t obtained by forking the simulator at t^* . Row (d) shows the locality violation outside D_t ; under noise coupling, any signal there is attributable to a locality violation rather than to sampling stochasticity. Row (e) repeats row (d) with the noise coupling removed, where the same measurement is no longer possible per sample.